

The Interview-Compatible Recommender: A Measuring Instrument for Cold-Start Conversational Recommendation

Thesis Chapter A (v2)

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Abstract

Before one can study *how* to interview a cold-start user—which questions to ask, in what order—one needs a recommender that can *be* interviewed: a model that folds an arbitrary-length, mixed-channel set of answers into its belief and re-ranks a large catalogue, without per-user retraining and without trading away full-profile accuracy. We argue that this recommender is a *measuring instrument* for the entire elicitation-policy literature, and that its requirements are more demanding than any single published system meets. We state five requirements—**R1** state-of-the-art full-profile accuracy, **R2** arbitrary-length cold-start fold-in, **R3** channel-agnostic input (items, out-of-catalogue concepts, entities, and continuous directions through one interface), **R4** an uncertainty-native belief, and **R5** monotonicity (a truthful answer never lowers ranking quality)—and three acceptance gates **G0/G1/G2** that operationalise them. A survey of six model families shows the literature is *bifurcated*: the models that hit the full-profile bar (shallow linear/autoencoder recommenders such as EASE [20] and RecVAE [19]) are item-only point estimators, while the models with an uncertainty-native, multi-channel belief (soft-attribute Gaussian elicitation [3], unified item+attribute bandits [11], conjugate critique folds [23]) do not report full-profile accuracy at the R1 bar. We are aware of no published system that occupies the conjunction of all five. The instrument we propose to close this gap is a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian belief layer over its latent, into which every channel folds as a linear observation; three of its input legs—signed/graded values, out-of-catalogue concepts, and continuous direction tokens—are architecturally inexpressible in the item-indicator basis of the R1-bar models. This chapter contributes the requirement framework, the taxonomy and gap analysis, the instrument’s design, and the experimental protocol (a metric-certification bridge to the published ML-20M numbers, then G0/G1/G2 on the ML-25M harness). Our two provenanced in-house anchors—a set-encoder tower at full/tail NDCG@10 0.4946/0.3372 and a RecVAE-class $d=512$ ruler at 0.4998/0.3443 (both ML-25M, not yet bridged to the published ML-20M scale)—are reported as such; all downstream instrument results are marked **[TODO:]** pending the certified run.

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1 Introduction: the recommender as a measuring instrument

A *cold-start* conversational recommender must elicit taste from a short interview and then rank a large catalogue. Two questions are entangled: *what to ask* (the elicitation policy) and *how to turn answers into recommendations* (the recommender). The elicitation-policy literature is large and active, but almost every comparison in it is confounded: a policy’s apparent value depends on the recommender it drives, and reported gains routinely conflate a better *question-selection strategy* with a better *answer-ingestion model*. Before we can measure elicitation strategies in isolation, we need to fix the recommender—and not just any recommender, but one strong and flexible enough that a good interview is not wasted on it. We call such a recommender an *interview-compatible instrument*.

Five requirements. An instrument fit for measuring conversational elicitation must simultaneously satisfy:

- R1 — full-profile SOTA.** On a full interaction history it must rank at the level of the strongest shallow collaborative-filtering baselines (EASE [20], RecVAE [19]), so that elicitation is measured against a recommender nobody can dismiss as weak [6].
- R2 — any-length cold-start fold-in.** It must ingest an evidence set of *any* cardinality, from 0 (fully cold) upward, without per-user retraining.
- R3 — channel-agnostic input.** Items, out-of-catalogue concepts, entities, and free text (and, for the continuous policy, direction tokens) must all fold through *one* interface as arbitrary embeddings, not a fixed categorical schema.
- R4 — uncertainty-native belief.** It must carry an explicit covariance over the user belief, not just a point estimate, so that a policy can act on *what it does not yet know*.
- R5 — monotonicity.** A truthful answer must never *lower* ranking quality; adding evidence should, in expectation, only help.

The bifurcation observation. The central empirical fact this chapter documents is that R1 and {R3, R4, R5} are *anti-correlated across the published literature*. The models that hit the full-profile bar—closed-form linear autoencoders (EASE [20], EDLAE [21]), amortised VAEs (Mult-VAE [12], RecVAE [19]), and training-free graph filters (GF-CF [18], BSPM [4], Turbo-CF [16])—are *item-only point estimators* on a one-hot indicator basis. The models with an uncertainty-native, multi-channel

belief—soft-attribute Gaussian elicitation [3], item+attribute Thompson bandits [11], conjugate critique folds over a frozen factorisation [23], and per-item Bayesian NL elicitation [1]—do *not* report full-profile accuracy at the R1 bar. We call these the *accuracy corner* and the *belief corner*. No published system we are aware of occupies both (§4).

Contributions. This chapter is the instrument paper of the thesis. It does not claim a new elicitation policy (that is the companion method chapter); it claims the *recommender that makes policy measurement possible*. Concretely:

1. **The conjunction as the contribution.** We frame $R1 \wedge R2 \wedge R3 \wedge R4 \wedge R5$ as a single requirement no published system meets, and give a six-family taxonomy and per-near-miss gap analysis (§3, §4) that locates exactly which requirement each candidate fails. We say we are *aware of none*, never that there *is* none.
2. **Three architecturally-inexpressible legs.** We isolate the three input channels that are impossible (not merely untuned) in the item-indicator basis of every R1-bar model: (i) signed/graded per-entity values, (ii) out-of-catalogue concepts, and (iii) continuous direction tokens. These are the narrow, defensible novelty legs of the channel-agnostic interface (§4, §5).
3. **The instrument.** A design for a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian precision-accumulator belief layer over its latent, into which every channel folds as a linear observation (§5). The frozen-tower+exact-conjugacy shape is the one mechanism with a principled monotonicity story [23], and it is what separates us from the co-trained, approximate-posterior near-miss of Bıyık et al. [3].
4. **G0/G1/G2 as a reusable acceptance protocol.** We define three gates—full-strength preserved, posterior strictly shrinks per question, and cold-start per-question NDCG rises on *full and tail* versus random—and propose them as a portable certificate any interview-compatible recommender should have to pass (§2, §6).

We are careful throughout to distinguish what is *new* (the conjunction; the frozen-certified-tower wedge; the G0–G2 protocol on a SOTA tower) from what is *pre-empted* and must be cited (attributes-as-directions plus a Bayesian fold [3]; unified item+attribute belief [11]; conjugate folds over a frozen factorisation [23]; permutation-invariant set-encoders with information-gain acquisition [15]; value-carrying set-input cold-start recommenders [13]; the non-degradation-of-information principle, Good 1967 / Blackwell). “Uncertainty-native recommender” alone is a crowded 2024–25 area [2] and is *not* claimed as novel.

2 Requirements and gates

We now state the requirements and gates formally. Let the catalogue have n items and let the frozen tower expose a d -dimensional latent in which each item i has a factor row $\mathbf{Q}_i \in \mathbb{R}^d$. An *evidence set* after t questions is a set of observations $\mathcal{E}_t = \{(\phi_j, y_j)\}_{j=1}^t$, where $\phi_j \in \mathbb{R}^d$ is the latent direction of the queried entity (an item factor, a concept centroid, an entity embedding, or a continuous direction token) and $y_j \in \mathbb{R}$ is the (possibly graded, possibly signed) answer. The instrument maintains a belief $b(\mathcal{E}_t) = (\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ over the user latent $\mathbf{u} \in \mathbb{R}^d$ and scores item i by a fixed read-out $s_i(\boldsymbol{\mu}_t)$ of the base tower.

Requirements (formal).

- R1.** With $\mathcal{E}$ equal to a user’s full history, $\text{NDCG@10}(s(\boldsymbol{\mu}(\mathcal{E})))$ ties the strongest shallow baseline on a shared harness (Def. 1).
- R2.** $b(\mathcal{E}_t)$ is defined and computable for every $t \geq 0$ with no gradient step at inference; $t=0$ yields the prior belief (a non-personalised ranker).
- R3.** The update from $\mathcal{E}_t$ to $\mathcal{E}_{t+1}$ depends on the new observation only through the pair $(\boldsymbol{\phi}_{t+1}, y_{t+1}) \in \mathbb{R}^d \times \mathbb{R}$; the operator is identical for every channel, so any entity with an embedding is a legal observation—including entities absent from the catalogue.
- R4.** $\boldsymbol{\Sigma}_t$ is an explicit, queryable covariance; a policy may read $\boldsymbol{\Sigma}_t$ to choose the next question.
- R5.** For a truthful answer y_{t+1} , $\mathbb{E}[\text{NDCG@10}(s(\boldsymbol{\mu}_{t+1}))] \geq \text{NDCG@10}(s(\boldsymbol{\mu}_t))$. We treat R5 as an *empirical* property to be demonstrated per step, not a theorem for a learned ranker (§7).

Definition 1 (Ties SOTA). *A recommender ties a baseline on a metric if their scores are within the paired per-user bootstrap confidence interval on one shared harness (same users, items, split, metric depth). A number computed on a different metric/split (e.g. NDCG@10 on ML-25M vs. NDCG@100 on the ML-20M strong-generalisation split) is not a tie until bridged (§6.2).*

Gates (acceptance protocol). The instrument is accepted only if it passes all three:

- G0 — full strength preserved.** Evaluating on the belief mean with the full profile reproduces the base tower’s full-profile NDCG@10 (full *and* tail), i.e. the belief layer adds no full-profile regression. This certifies R1 survives the addition of the belief machinery.
- G1 — usable posterior.** The posterior strictly shrinks per truthful question: $\det \boldsymbol{\Sigma}_{t+1} < \det \boldsymbol{\Sigma}_t$ (and the shrinkage is directed along the queried $\boldsymbol{\phi}_{t+1}$). This certifies R4.
- G2 — cold-start curve rises on full and tail.** Starting cold ($t=0$), per-question NDCG@10 rises with t on *both* the full catalogue and the long tail, and beats a random-question control at every budget with a paired per-user bootstrap CI excluding zero. This is the operational form of $\text{R2} \wedge \text{R5}$ and the headline instrument claim.

We report *both* full and tail NDCG@10 for every gate (a project reporting rule: full-only or tail-only hides the effect, which historically has been much larger on the tail than on the full catalogue). All gate results in this chapter are **[TODO:]** pending the certified run of §6.

3 Related work: a six-family taxonomy

We organise interview-relevant recommenders into six families and score each against R1–R5 (Table 1). The reading is consistent across families: no family is strong on both the accuracy axis (R1) and the belief-channel axis (R3–R5). Figure 1 plots the same fact as a qualitative gap map.

The accuracy corner (families ii–iii). On the canonical ML-20M strong-generalisation split (10k validation / 10k test held-out users, NDCG@100), the full-profile frontier has been *flat since 2019–2020* and is held by shallow models, not deep nets: EASE [20] at NDCG@100 0.420, MultVAE [12] at 0.426, RecVAE [19] at 0.442 (top of this split), with EDLAE [21] in the same class; SLIM (0.401) and WMF/iALS (0.386) are the classic floors. Dacrema et al. [6] is the binding warning: most deeper “SOTA” gains vanish on this exact split. The graph-filter leaders (GF-CF [18], BSPM [4], Turbo-CF [16]) are the 2023–24 speed+accuracy front on Gowalla/Yelp/Amazon but do not report this ML-20M split, so they are admissible to our accuracy tier only if re-run on our

harness. All of family (ii)/(iii) are item-only: their input is a one-hot (or set-indicator) interaction vector, which *cannot encode* a signed value, an out-of-catalogue concept, or a continuous direction—R3 is an architecture property here, not a tuning gap.

The belief corner (family iv, and vi). Bıyık et al. [3] maintain a multivariate-Gaussian belief over a user embedding and treat both item comparisons and “soft attributes” (e.g. “scarier”) as directions (concept-activation vectors) in the item space, selecting queries by expected value of information—the nearest thing to R2+R3+R4 together. ConTS [11] unifies items and attributes as Thompson-sampling arms with a posterior over a cold-start user. BCIE [23] folds critiques into a *frozen* factorisation by exact conjugate-Gaussian updates—the closest published analogue of our belief mechanism. PEBOL [1] keeps a per-item Beta posterior updated by LLM-extracted aspects and NLI scoring, reporting cold-start MRR@10 0.27 versus a 0.17 baseline. None of these reports full-profile accuracy at the EASE/RecVAE bar; each is a belief layer, not a certified tower.

The set-encoder and cold-start-NP precedents (families i, iii). EDDI/Partial-VAE [15] is the permutation-invariant set-encoder over an arbitrary observed subset with information-gain acquisition—exactly R2’s architecture, but demonstrated outside recommendation and without a full-profile CF result or an attribute/concept channel. TaNP [13] gives a neural-process, stochastic-process uncertainty over a value-carrying set input for cold-start—the R2+R4 shape, but not at full-profile SOTA nor channel-agnostic. These are cited as the mechanism precedents our design instantiates, not as rivals for the conjunction.

The gap map. Figure 1 places the families on two qualitative axes: *interview-compatibility* (channels + uncertainty + monotonicity, i.e. R3–R5) on the x -axis, and *full-profile accuracy* (R1) on the y -axis. The accuracy-corner systems sit top-left; the belief-corner systems sit lower-right; the target sits top-right, empty.

4 Gap analysis

The verdict. No single published system satisfies all of $\{R1 \text{ SOTA-full-profile} \wedge R2 \text{ any-length set} \wedge R3 \text{ channel-agnostic via arbitrary embeddings} \wedge R4 \text{ uncertainty-native} \wedge R5 \text{ monotone}\}$. The binding near-miss is Bıyık et al. [3]: it fails R1 (a co-trained item space, not a frozen certified RecVAE-class tower, and no full-profile benchmark) and gives only an approximate, “generally not Gaussian” posterior; every other candidate fails R3 or R4 outright. **We are aware of none** that occupies the conjunction—and we state it exactly that way, never as “there is none.”

Per-near-miss failure analysis.

- **Bıyık et al. 2023 [3]** — has R2 + R3 (items and soft attributes as directions) + R4 (Gaussian belief). *Fails R1*: the item space is co-trained with the belief model, not a certified full-profile CF tower, and accuracy is reported on their elicitation task, not the EASE/RecVAE bar. R3 is *attribute directions*, not arbitrary embeddings / open concepts / continuous query tokens. R5 is approximate (sampled EVOI over a non-Gaussian posterior). This is the central threat, and our wedge is exactly (a) frozen certified tower vs. co-trained, (b) exact conjugacy vs. approximate posterior, (c) arbitrary / open / continuous tokens vs. a fixed attribute set.

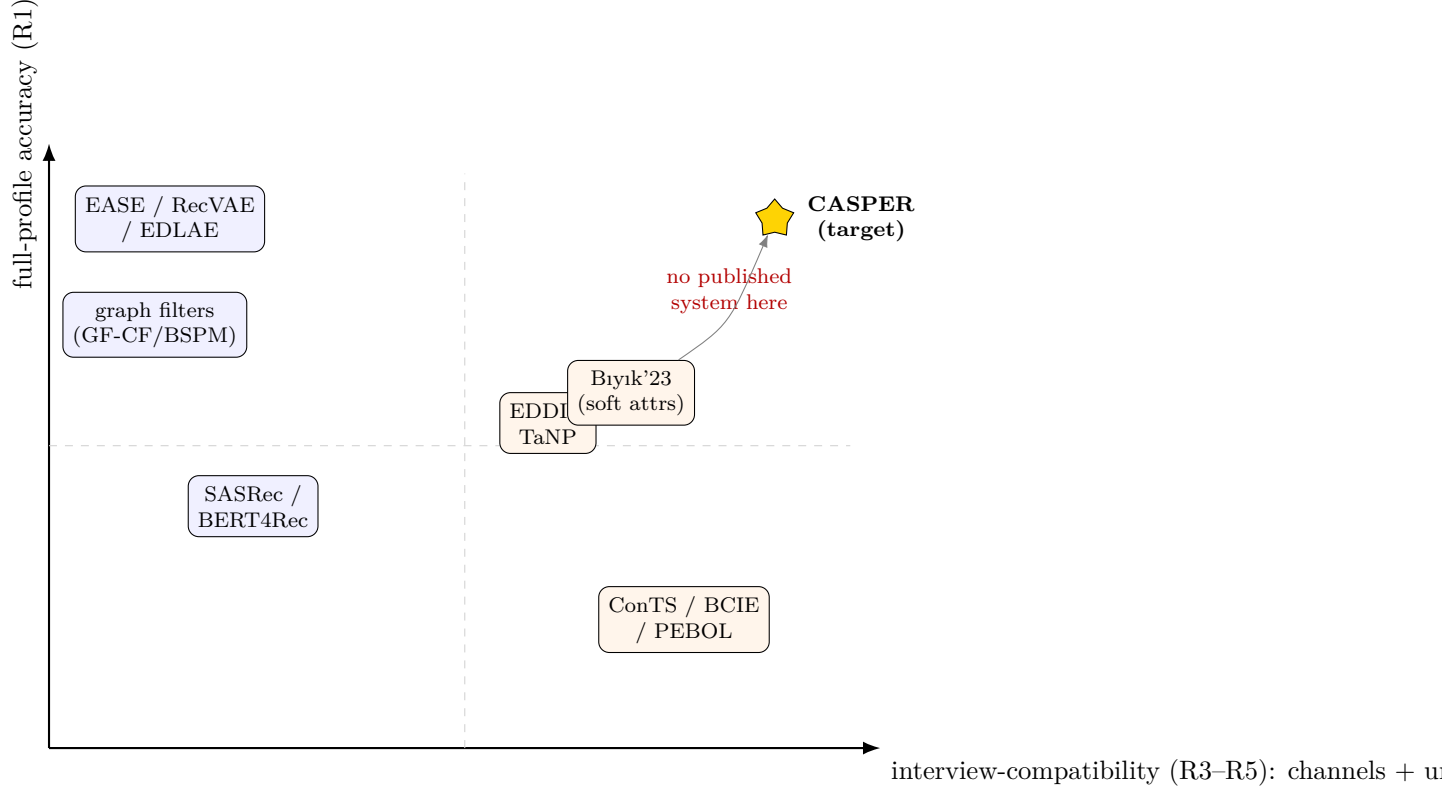


Figure 1: **The gap map (qualitative).** x -axis: interview-compatibility (channel-agnostic input, an uncertainty-native belief, and monotonicity—R3–R5). y -axis: full-profile accuracy (R1). Positions are *qualitative* and reflect our reading of each system’s published claims, not measured coordinates; they are meant to show the *shape* of the bifurcation, not exact rankings. The accuracy corner (EASE/RecVAE/EDLAE [20, 19, 21], graph filters [18, 4], sequence models [9, 22]) is top-left; the belief corner (EDDI [15]/TaNP [13], Biyık [3], ConTS [11]/BCIE [23]/PEBOL [1]) is lower-right. The top-right target is empty: we are aware of no published system there.

- **ConTS** [11] — R2 + R4 + a unified item/attribute action space. *Fails R3* (a fixed categorical attribute schema; no arbitrary embedding, no continuous or open token) and *R1* (a bandit over arms, not a full-profile encoder).
- **EDDI / Partial-VAE** [15] — R2 + R4 (information-gain) + set input. *Fails R1* (not a recommendation model / no CF SOTA) and *R3* (no attribute or concept channel as demonstrated).
- **RecVAE / EASE** [19, 20] — *R1 yes*. *Fail R3* (an item-only one-hot / indicator basis—cannot express a concept, a signed value, or a continuous direction) and *R4* (point estimate; RecVAE’s latent posterior is item-only and is not used as an actionable belief).
- **PEBOL** [1] — R4 (per-item Beta) + R3 (LLM aspects) + strong cold-start. *Fails R1* (no full-profile CF ranking) and its R4 is *per item*, not a shared user covariance; no monotonicity guarantee.
- **SASRec / BERT4Rec** [9, 22] — R2 (re-encode the sequence). *Fail R3, R4, R5* (item-only, order-sensitive, point estimate, non-monotone—a later token can reorder everything). Token input is trivially satisfied but is *not* value-carrying or mixed-type, so “interview = tokens” does

not hold; and BERT4Rec’s fold-in advantage carries a replicability caveat [17].

The three inexpressible legs. Three input channels are *architecturally impossible* (not merely untuned) in the item-indicator basis of every R1-bar model: (i) graded/signed per-entity values (the basis has no sign and no magnitude—a “dislike” is not representable as a negative one-hot without changing the likelihood), (ii) out-of-catalogue concepts (there is no column for an entity absent from the catalogue), and (iii) arbitrary continuous direction tokens. Folding these as *linear observations in a frozen latent* is the one leg of the conjunction we find nowhere in the literature.

Pre-empted—cited, never claimed as first. We must credit, and do not claim priority over: attribute-answers-as-directions plus a Bayesian fold [3]; unified item+attribute belief and value-of-information [3, 11]; conjugate-Gaussian folds over a frozen factorisation [23]; permutation-invariant set-encoders with information-gain acquisition [15]; value-carrying set-input cold-start recommenders [13]; the non-degradation-of-information principle (Good 1967 / Blackwell); and the now-crowded “uncertainty-native recommender” area [2]. Our defensible novelty is only the *conjunction* of all five requirements in one frozen certified tower, plus the three inexpressible legs and the G0–G2 protocol.

5 The instrument

The design follows directly from the gap analysis: use the shallow frozen tower that already holds the accuracy corner (R1, and the flat-since-2019 frontier means no deep-net detour is warranted), and add the one belief mechanism with a principled monotonicity story—exact linear-Gaussian conjugacy—over its frozen latent (R4, R5), with a channel-agnostic linear-observation interface (R3). This section is written at design level; implementation details and all numbers are **[TODO:]**.

5.1 Frozen shallow-SOTA tower

The substrate is a certified shallow recommender—a RecVAE-class amortised encoder or a closed-form autoencoder [19, 20]—trained once on the full catalogue and then *frozen*. It exposes (a) item factor rows $\mathbf{Q}_i \in \mathbb{R}^d$, (b) a decoder read-out $s_i(\mathbf{u}) = \mathbf{Q}_i^\top \mathbf{u} + \beta_i$ with a learned bias β_i , and (c) the encoder that maps a full interaction set to a latent. Freezing is the wedge against Bıyık [3] (co-trained) and the guarantee that G0 (full strength preserved) is a property of the substrate, not of the belief layer. We deliberately do *not* extend the linear basis with concept columns (that stays in the item-indicator world and fails R3), nor learn the uncertainty head (no monotonicity story, and retraining risks G0). **[TODO: Select and certify the frozen tower (RecVAE-d512 ruler candidate; report the certified full/tail NDCG@10 and the bridge number of §6.2).]**

5.2 Analytic conjugate-Gaussian belief layer

Over the frozen latent we place a Gaussian belief on the user vector, $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, and model each answer as a noisy linear read-out along the queried direction:

$$y_j = \boldsymbol{\phi}_j^\top \mathbf{u} + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \quad (1)$$

Because the observation is linear-Gaussian, the posterior after $\mathcal{E}_t$ is Gaussian in closed form; maintaining it in *precision* form (a precision accumulator) makes the any-length fold-in a single running

sum:

$$\mathbf{\Lambda}_t = \mathbf{\Lambda}_0 + \sum_{j=1}^t \sigma_j^{-2} \phi_j \phi_j^\top, \quad \mathbf{\Lambda}_t \boldsymbol{\mu}_t = \mathbf{\Lambda}_0 \boldsymbol{\mu}_0 + \sum_{j=1}^t \sigma_j^{-2} y_j \phi_j, \quad \boldsymbol{\Sigma}_t = \mathbf{\Lambda}_t^{-1}. \quad (2)$$

Equation (2) is exactly the mechanism BCIE [23] uses for critiques over a frozen factorisation; our contribution is doing it over a *certified full-profile SOTA* tower with channel-agnostic tokens, and certifying the G0–G2 protocol on it. Three properties fall out by construction: it is a permutation-invariant *set* operation defined for every $t \geq 0$ (R2); each update adds a rank-1 term along ϕ_{t+1} , so $\mathbf{\Lambda}_t$ only grows and $\det \boldsymbol{\Sigma}_t$ only shrinks (G1); and conjugate updates are Blackwell/Good-1967 monotone in *expected utility*. The last is why R5 is plausible, but—because the tower’s dot-product ranker does not act optimally under the belief—per-question NDCG monotonicity is *not* automatic and must be shown empirically (§7). **[TODO: Fix $\mathbf{\Lambda}_0, \boldsymbol{\mu}_0$ (prior from the tower), the per-channel noise σ_j , and the confidence/temperature schedule; verify G0 holds at the belief mean on the full profile.]**

5.3 Channel-agnostic tokens as linear observations

Every channel supplies a direction $\phi_j \in \mathbb{R}^d$ and a value $y_j \in \mathbb{R}$ to Eq. (2); the update is identical across channels (R3):

- **Items.** $\phi_j = \mathbf{Q}_i$ (the frozen item factor). A graded or signed answer sets y_j ; a dislike is a negative y_j along the same direction—representable here precisely because the observation model (1) has a sign and a magnitude, unlike a one-hot indicator.
- **Out-of-catalogue concepts.** ϕ_j is a concept direction in the latent (e.g. a centroid of member factors, or a text-adaptor image of the concept), folded exactly like an item—no catalogue column required.
- **Entities / free text.** An embedding is projected into the latent by a fixed adapter and folded identically.
- **Continuous direction tokens.** A policy may emit any $\phi_j \in \mathbb{R}^d$ directly (the continuous-action interface the companion chapter uses); the belief update does not distinguish it from a catalogue entity.

The signed/graded, out-of-catalogue, and continuous legs are the three channels inexpressible in the R1-bar basis (§4). **[TODO: Specify the concept-direction and text-adaptor constructions, the polarity/gradation encoding, and the acceptance tests (a like/dislike flip reverses the queried neighbourhood; a concept fold lifts its member items; monotone in t).]**

6 Experimental design

6.1 Baseline bank

The baseline bank (Table 2) is three tiers: classic floors that every curve must clear (Dacrema et al. [6] demands them), the full-profile SOTA bar that decides G0, and the elicitation-compatible rivals that decide G1/G2. Reported anchors are on the systems’ *own* splits and are not comparable to ours until bridged (§6.2); we print them only to show what “ties SOTA” will mean.

6.2 Evaluation protocol

(i) **Certification bridge.** Our in-house anchors are on a *different* ruler than the published bar, and we do not conflate them. Concretely, our set-encoder tower reaches full/tail NDCG@10 0.4946/0.3372 and our RecVAE-class $d=512$ ruler reaches 0.4998/0.3443, *both on ML-25M at NDCG@10*—whereas the published EASE/RecVAE numbers are NDCG@100 on the ML-20M strong-generalisation split. These are not comparable, and no “ties SOTA” claim is made until the bridge is run. The bridge has two steps: (1) *reproduce* the published EASE and Mult-VAE/RecVAE numbers on the ML-20M Liang split (NDCG@100, Recall@20/50) in our own code, snapping to the reported values exactly [20, 12, 19]; (2) *port* the certified tower to the ML-25M harness and report full/tail NDCG@10 there. We keep ML-20M in the loop despite being older because the Liang 2018 split is a protocol lock-in the whole full-profile literature reports on; ML-25M is the newer corpus and, crucially, carries the genome tags the concept channel needs, so both datasets are required—one for comparability, one for the channels. [TODO: Bridge table: reproduced ML-20M NDCG@100/Recall vs. published; ported ML-25M full/tail NDCG@10.]

(ii) **G0 — full-profile on ML-25M, items-only.** On the baselines’ native input (items only), evaluate full and tail NDCG@10 on the full ML-25M catalogue with a paired per-user bootstrap CI, confirming the instrument’s belief mean reproduces the frozen tower and ties EASE/RecVAE on the shared harness. [TODO: G0 table (full & tail NDCG@10, all baselines, CIs).]

(iii) **G2 — per-question curves.** Cold-start per-question NDCG@10 curves (full and tail) versus a random-question control. Items are foldable by all baselines; mixed channels (concepts, signed values, continuous tokens) are foldable only by our instrument, which is the point of the comparison—the rivals cap out at what an item-only interview can carry. Symmetric access is enforced where a baseline *can* fold a channel; where it structurally cannot, that is reported as an architecture limit, not hidden. [TODO: G2 curves for items-only (all methods) and mixed-channel (ours), full & tail, with CIs and the random control; G1 posterior-shrinkage diagnostic.]

(iv) **Concept-as-pseudo-item ablation.** To pre-empt “just add genre columns,” we fold a concept into EASE/RecVAE as the centroid-of-members pseudo-item and compare against our concept channel. The expected finding is that a pseudo-item column is a lossy special case (it cannot carry sign, magnitude, or an out-of-catalogue direction), but we run it rather than assert it. [TODO: Ablation table: concept-as-pseudo-item fold into EASE/RecVAE vs. our concept channel, full & tail.]

Metrics. Full *and* tail NDCG@10 on ML-25M (the project reporting rule, scored with the decoder’s learned bias, not a log-count popularity floor), plus NDCG@100 (and Recall@20/50) on the ML-20M bridge split for comparability with the published anchors. Every headline delta carries a paired per-user bootstrap CI on one shared harness. All result cells are [TODO:].

7 Limitations and threats to validity

- **Metric non-comparability until the bridge is run.** Our 0.4946/0.3372 and 0.4998/0.3443 are NDCG@10 on ML-25M; the published bar is NDCG@100 on ML-20M. Until §6.2 snaps our code to the published numbers, *no* “ties SOTA” claim is licensed—the numbers live on different rulers.

- **Monotonicity is empirical, not a theorem.** Conjugate-Gaussian updates are monotone in expected utility (Good 1967 / Blackwell) *only in expectation*, and only if the ranker acts optimally under the belief—which a fixed dot-product / NDCG ranker does not. R5 (and G2’s rising curve) is therefore a *measured* property with a per-question CI, not a guarantee; a truthful answer can in principle still lower NDCG at a given step, and we report any such step rather than suppress it.
- **Simulator-based evaluation pending a human study.** G1/G2 are measured under a stated answer model (a simulator), which shares structure with the instrument’s own latent; this risks circularity (the answer, the belief, and the metric are all functions of the same space). We add answer-noise ablations and a simulator-alignment limitation to every gate, and treat a ≥ 1 human round-trip study as the load-bearing external validation—not yet run.
- **BERT4Rec replicability.** The sequence baseline’s fold-in advantage is subject to a documented replicability caveat [17]; we tune it to the published protocol and flag any residual gap rather than treat a single run as authoritative.
- **Graph-filter admissibility.** GF-CF/BSPM/Turbo-CF [18, 4, 16] do not publish the ML-20M strong-generalisation NDCG; they enter Tier 2 only if re-run on our split, and are otherwise reported as an accuracy ceiling on other datasets, not as a bridged tie.
- **Absence of evidence.** The conjunction claim is “we are aware of none,” bounded by our survey; a set-encoder / deep-set recommender that hits RecVAE-class full-profile accuracy *and* takes value-carrying mixed-type tokens would partly pre-empt $R1 \wedge R2 \wedge R3$. Our searches found no such reported system, which is absence of evidence, not proof of absence.

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Family	Representatives	R1	R2	R3	R4	R5	One-line verdict
(i) retrain / meta-learn per user	MeLU, TaNP [13], CDRNP	~	✓	✗	~	✗	R2/R4 precedent; not full-profile SOTA, not channel-open.
(ii) fold-in linear / closed-form	EASE [20], SLIM, GF-CF [18], BSPM [4], Turbo-CF [16]	✓	✓	✗	✗	~	The full-profile bar (G0); item-indicator basis $\Rightarrow$ R3 impossible.
(iii) amortised inference encoders	Mult-VAE [12], RecVAE [19], EDLAE [21], EDDI [15]	✓	✓	~	~	✗	The architecture we live in; VAEs item-only, EDDI right shape/wrong domain.
(iv) Bayesian belief over item space	Bıyık [3], ConTS [11], BCIE [23], ICF [24]	✗	✓	~	✓	~	The R4/R5 heart; belief layers, not full-profile towers.
(v) sequence models	SASRec [9], BERT4Rec [22]	~	✓	✗	✗	✗	Interview $\neq$ ordered session; tokens item-only, order-sensitive.
(vi) LLM-as-recommender / CRS	PEBOL [1], GATE [10], LLM-CRS [8]	✗	✓	✓	~	✗	Best R3 (open text); weakest R1 (weak collaborative signal).
CASPER (target)	this chapter	✓	✓	✓	✓	✓	The conjunction; no published system occupies it.

Table 1: Six-family taxonomy scored against R1–R5. ✓ met, ~ partial, ✗ not met. Scores are our reading of each family’s *published* claims and are qualitative; the last row is a target, not a result. R1 and {R3,R4,R5} are anti-correlated: families (ii)/(iii) hold the accuracy corner, family (iv) the belief corner, and none spans both. Adapted from the landscape analysis (c2).

Baseline	Type	Reported anchor	Why the slot
<i>Tier 1 — classic floors (must clear)</i>			
Most-Popular	non-personalised	popb floor (our harness)	the “did we learn anything” line; G2 must beat it cold.
Item-kNN (cosine)	neighbourhood	ML-20M NDCG@100 ~ 0.36 class	instant fold-in; Cremonesi-era strong-simple [5].
PureSVD / iALS-WMF	linear MF fold-in	WMF ML-20M NDCG@100 0.386	cheap fold-in reference.
SLIM	sparse linear item-item	ML-20M NDCG@100 0.401	direct EASE ancestor.
<i>Tier 2 — full-profile SOTA (the R1 / G0 bar)</i>			
EASE [20]	closed-form item-item	NDCG@100 0.420	the bar to tie; item-only shows the R3 gap.
RecVAE [19]	amortised VAE	NDCG@100 0.442	top of the ML-20M split; our <i>d512</i> ruler class.
Mult-VAE DAE [12]	/ VAE / DAE	NDCG@100 0.426 / 0.419	canonical amortised anchor.
EDLAE [21]	denoising higher-order EASE	≈ RecVAE class	confirms the flat frontier.
GF-CF / BSPM / Turbo-CF [18, 4, 16]	closed-form graph filter	SOTA on Gowalla/Yelp (ML-20M not reported)	admit only if re-run on our split.
SASRec / BERT4Rec [9, 22]	self-attentive sequence	strong next-item (not this split)	sequence fold-in; watch replicability [17].
<i>Tier 3 — elicitation-compatible rivals (decide G1/G2)</i>			
EDDI / Partial-VAE [15]	set-encoder + info-gain	non-recsys (UCI/MNIST)	the R2 architecture; myopic info-gain baseline.
Bıyık attributes [3]	soft-Gaussian belief + EVOI	own elicitation task	the central threat; frozen-vs-co-trained wedge.
ConTS [11]	Thompson bandit, item+attr arms	cold-start CRS metrics	unified-space + uncertainty; fixed schema.
BCIE [23]	conjugate-Gaussian critique fold	critique metrics	exact-conjugacy-over-frozen precedent.
PEBOL [1]	Bayesian-opt NL-PE (LLM)	MRR@10 0.27 vs 0.17	load-bearing PE baseline; per-item Beta.
Golbandi tree [7]	RMSE decision-tree interview	Netflix RMSE	the static-tree interview comparator.
RBMF / Functional-MF [14, 25]	MF-embedding seeds	elicitation RMSE	interview-seed line; cite-and-differentiate.

Table 2: Baseline bank. Anchors are on each system’s own split/metric and are *not* comparable to ours until the bridge (§6.2) is run; G0 is decided against EASE+RecVAE, G1/G2 against the Tier-3 rivals plus a random-question control. Any deep-net “SOTA” not snapping to Tier-2 numbers on our split is inadmissible [6].